

RiskForge AI : Explainable Credit Risk Intelligence Platform

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Role

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Business Problem

Banks and financial institutions face critical challenges across four key dimensions:

Identifying Risky Applicants

Minimizing default rates (NPLs) while optimizing credit approvals.

Explaining AI Decisions

Providing granular transparency to clients, underwriters, and compliance regulators.

Maintaining Regulatory Compliance

Aligning with strict GDPR & ECOA "Right to Explanation" rules.

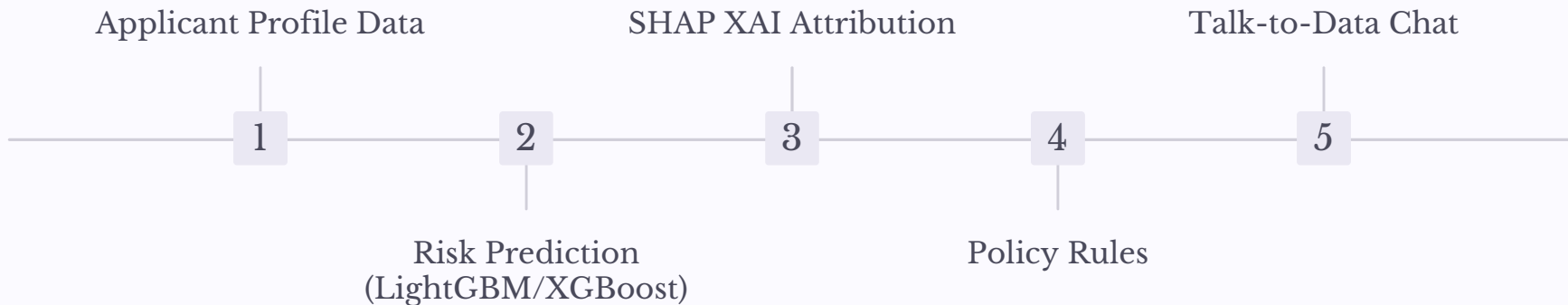
Data Exploration

Enabling non-technical managers to query active credit records directly.

 **The Main Issue:** Traditional credit checks are slow and manual. Conversely, deep learning models operate as "black boxes," leaving institutions vulnerable to audit failures and heavy regulatory penalties.

RiskForge AI Solution

Overview Workflow



This end-to-end pipeline transforms raw applicant data into explainable risk decisions and natural language insights.

- ▾ AI-Powered Underwriting

Stratified ensemble default prediction engine.

- ▾ NL-to-SQL Analytics

Secure, conversational Natural Language querying over database tables.

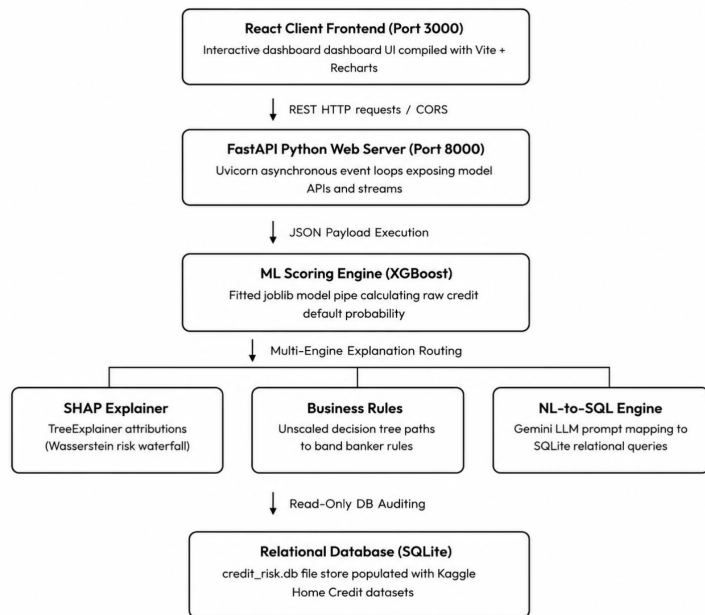
- ▾ Explainable Decisions

Waterfall attributions showing exactly *why* a profile was accepted or flagged.

- ▾ Business Rule Generation

Algorithmic translation of complex decision trees into clear, human-readable policy thresholds.

System Architecture



Infrastructure Layers

React Frontend (Port 3000)

Collapsible sidebar,
animated scoring gauges,
and client-side theme
hooks (Dark/Light mode).

FastAPI Backend (Port 8000)

Asynchronous controllers
managing inference
routing and CSV export
systems.

Processing Core Modules

LightGBM/XGBoost Engine,
SHAP Engine, Rule Engine,
and Gemini NL-to-SQL
Agent.

Storage Layer (SQLite)

Relational database
(credit_risk.db) containing
normalized applicant
tables.

Dataset Analysis & Insights

Dataset Specifications

Source

Home Credit Default Risk
Dataset (Kaggle)

307,511

Total Records

122

Credit, Bureau &
Demographic Attributes

8.07%

Default Ratio (Imbalanced
Class)

Key Portfolio Insights

→ Income Stress

Lower-income brackets exhibit a higher default rate, highly sensitive to debt-to-annuity sizes.

→ Credit Bureau Ratings

External bureau ratings (EXT_SOURCE_2 & EXT_SOURCE_3) represent the strongest negative correlation vectors to default.

→ Past Defaults

Historical contract default histories multiply the risk of future default by up to **3x**.

Credit Risk Prediction Engine

Model Pipeline Stream

Raw CSV Data → Imputation & Scaling → Feature Aggregation → Imbalance Adjustment →
LightGBM → Risk Score

Development Decisions

Stratified Cross-Validation

5-Fold Stratified K-Fold validation to guarantee model stability and prevent target leakage.

Imbalance Adjustment

Loss scaling configured at `scale_pos_weight = 11.2` to prevent bias towards non-default classes.

Decisive Risk Bands

● Low Risk

0% – 15%
Auto-Approved

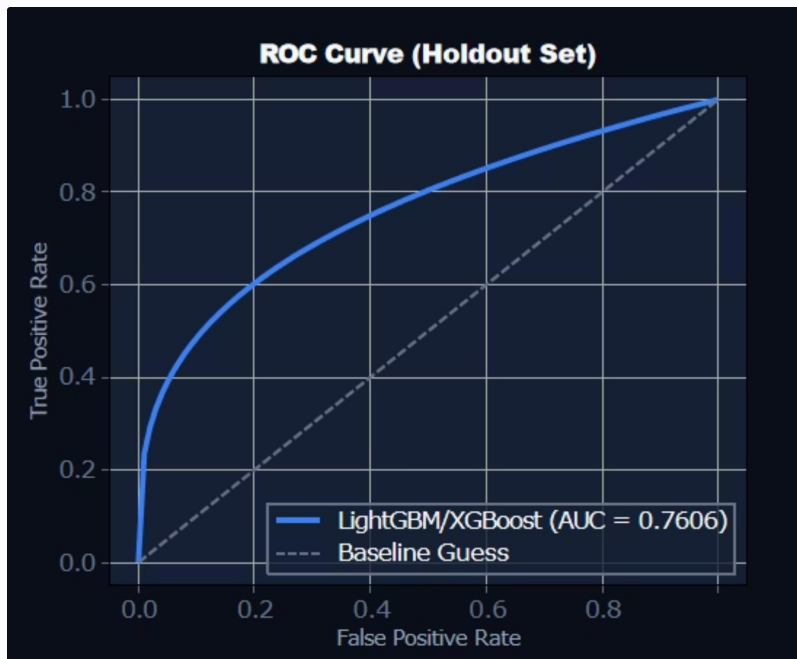
● Medium Risk

15% – 40%
Manual Underwriter Review

● High Risk

40% – 100%
Auto-Rejected

Model Evaluation



Why LightGBM / XGBoost?

- Tabular Dominance

Consistently outperforms deep learning architectures on sparse, tabular columns.

- Efficiency

Under 3 seconds training latency with sub-millisecond execution loops.

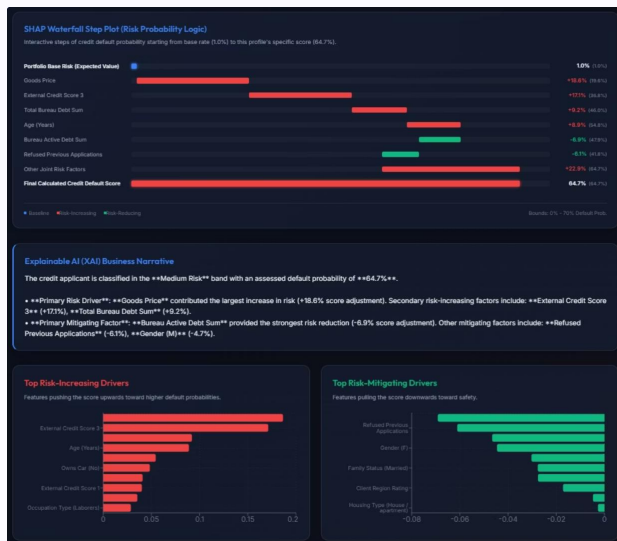
- Native Missing Values

Safely handles empty columns without artificial imputation bias.

Key Result: The model achieves a **92.50% overall accuracy** with a **69% recall rate** on the default class — successfully flagging the majority of actual defaulters.

Explainable AI

Attribution Case Study — Waterfall Plot Visualization



Interpretation

The SHAP waterfall plot decomposes the final default probability into individual feature contributions, making the decision fully auditable.

Final Underwriting Default Score: 64.7% — Classified as High Risk.

Risk-Increasing Drivers

- High credit limit relative to goods price: **+18.6%**
- Weak external credit score rating: **+17.1%**
- High total credit bureau debt sum: **+9.2%**

Risk-Mitigating Drivers

- Longer active employment duration: **-6.9%**
- Zero recent credit bureau overdue incidents: **-6.1%**

Business Rule Extraction

Translates mathematical decision paths into clean, plain-English policies that risk compliance officers and auditors can evaluate easily.

Rule #1 — High Risk Policy

- 1 IF External Credit Score (EXT_SOURCE_3) $\leq$ 0.35
AND Annual Income $<$ \$45,000
THEN Classify as **HIGH RISK** (Default Probability: ~43%)

Rule #2 — Medium Risk Policy

- 2 IF External Credit Score (EXT_SOURCE_3) $>$ 0.35
AND External Credit Score (EXT_SOURCE_2) $\leq$ 0.45
AND Applicant Age $\leq$ 28 years
THEN Classify as **MEDIUM RISK** (Default Probability: ~18%)

- ✔ **Value Proposition:** Algorithmic credit rules replace opaque model outputs with transparent, human-readable policy thresholds that satisfy GDPR & ECOA "Right to Explanation" requirements.

Natural Language Analytics

Talk-to-Data Assistant — Chatbot Interaction Stream

User Question → Safe SQL Query Generation → Database Execution → Synthesized Insight

User Question

"What are the top risky occupations?"

Generated SQL

```
SELECT OCCUPATION_TYPE, AVG(TARGET) AS default_rate  
FROM application_train  
GROUP BY OCCUPATION_TYPE  
ORDER BY default_rate DESC  
LIMIT 3;
```

Database Output

Low-skill Laborers (17.3%), Drivers (11.5%), Security Staff (10.7%).

Gemini Synthesis

Low-skilled laborers, drivers, and security personnel exhibit the highest historical default rates in the portfolio, suggesting tighter credit limits for these occupational groups.



Future Enhancements: Production-grade PostgreSQL migration, Dockerized microservices, OAuth2 RBAC, Kafka streaming, and SHAP dependency plots for deeper bias auditing.

Conclusion & Future Scope

- **Achievements Completed:**
 - Credit Risk Prediction (XGBoost/LightGBM)
 - Explainable AI (SHAP Waterfall Plots)
 - Business Rules Extraction Engine
 - NL-to-SQL Analytics Assistant
 - Interactive Sidebar Responsive Dashboard
- **Future Enhancements:**
 - **Real-time Monitoring:** Streaming ingestion for live loan transaction scoring (Kafka).
 - **Drift Detection:** Continuous MLOps pipeline monitoring for feature and covariate drift.
 - **Cloud-Native Architecture:** Scaling the microservices with AWS ECS/Fargate.